

Ethical Implication of Smart Devices in Homes and Businesses

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Introduction

Since the 2010's, smart home devices have been a common item in people's homes and their place of business. These devices include things like voice assistants like the Amazon Echo, smart speakers like Google Nest, and overall connected systems like the ones in modern day vehicles. These devices are so common because of their convenience, automation, and affordability. They have become so widespread because they connect to everyday objects through the internet so that they can exchange data to be autonomous. This autonomy makes the devices very user friendly. The affordability factor has really come into play because of the dropping prices in sensors and chips throughout the industry. These factors have led smart devices to be extremely common in today's modern home. According to a survey conducted by American Home Shield in 2024, 93% of American homeowners and renters have at least one smart device within their home (AHS). This number is astounding. It has become a widespread norm in the world, especially in America.

So why should we care? We should care because we are engineers. These devices were designed by engineers who rely on data collection, networking, and embedded systems to create a better world. We should care because even though these devices are meant for a good cause, there are situations where they are not. The main ethical concern of smart devices within homes and businesses is the fact that they collect and process large amounts of personal and private data. This brings up major controversy of drawing the line between convenience and innovation compared to privacy and user control. Devices that collect and store so much data require a company to process and handle the information. This leads to a case where a company could be collecting and using a person's data beyond what that person thinks they are using it for. This case raises ethical implications concerning data ownership, informed consent, corporate responsibility, and government interference which requires the use of ethical frameworks to come to the best solution possible (IEEE Digital Privacy).

Background

Smart home devices are designed to always be listening for words that "wake" them up. This means that they are always technically active, even if the user thinks that they are powered off. They also are designed to be self-learning. This means that the device can "learn the homeowner's schedules and make adjustments as needed" (Investopedia). Essentially, this makes it so the device stores a user's data so that they can use and adjust it as necessary. These devices can do things like send data to servers of third-party companies, store voice recordings, and use those recordings to improve the device's algorithms so the company can analyze the new data. To these various companies' credit, they do include sections in their terms of service that allow data usage on their end. However, the key issue is that smart device users are unaware of how much of their data is collected, how long the data is stored, and who actually has access to it.

Even though the user gives consent in the company's terms of service, they often don't read the actual document because they are far too long and complex.

Stakeholders

The stakeholders that are involved within this ethical issue have a lot of depth and are complicated. They include the consumers, the engineers and companies who manufacture and design the devices, third party companies who collect and use the data, and governments have a responsibility to protect its consumers.

Consumers

Consumers are the biggest stakeholder in this ethical dilemma. Overall, they want easy access to convenience and functionality. Also, they have a basic expectation of privacy in their homes and private spaces. Lastly, an everyday customer will most likely not know or understand the risks that they face when using these devices.

Engineers/Companies

The engineers and companies that design and manufacture these products have a big say in how the products are designed and how their systems are maintained. They also always want to improve the devices' performance by integrating them with artificial intelligence and by always updating their personalization abilities. Finally, the engineers and companies are always operating under pressure. This comes from the company wanting to make a profit and wanting to beat their competition within the industry.

Third Party Companies

Third party companies consist of advertising agencies and data analytics companies. These companies benefit directly from user data to formulate their advertising strategies and data trends to improve the customer's experience.

Governments

The governments that have constituents who use these smart devices are stakeholders in this ethical case. This is because they have a responsibility to protect consumers who are using smart devices. Governments need to dedicate their time and resources to balance innovation with regulation.

The impact of this ethical issue has implications all over the place. The use of smart devices within homes has grown exponentially in the past ten years. This makes it so these homes are increasingly reliant on systems that are data driven. With so much data being held, this can lead to data breaches that would have a big impact on society. Take Yahoo from 2013-2016 for example. A team of Russian hackers committed a cyberattack on Yahoo's database by using

backdoors and cookies to steal records from their user's accounts (UpGuard). This led to over 3 billion user accounts being exposed, fines being held, and 41 class-action lawsuits. As the reader can see, a data breach from a database that holds so much user data can be very detrimental to society. Data is a valuable asset that incentivizes hackers to get information and cybersecurity attacks to occur. Lastly, the social implications of these breaches lead to the expectation of changes in privacy for the user.

Methods

For this ethical case study, I will use the ethical frameworks of deontology and consequentialism to apply them towards my recommendations for the case.

Deontology

The main idea of the ethical framework of deontology is that a person's actions should be based on their individual duties, rights, and rules. This means that they are not focused on the outcome of a decision, but rather the moral obligations of the person making the decision. This framework ensures that individuals should never be treated as the means to an end.

This framework applies to this case in a few ways. First of all, users have a right to their own privacy, they have a right to informed consent, and they have a right to have autonomy over their own personal data. Furthermore, companies and engineers who design these products have a duty to be transparent with their product's operations, to avoid practices that can be used to be deceitful, and a duty to protect their user's private information. From the user's point of view, it is hard to give a truly informed user consent if the terms and conditions section of the product is extremely long and hard to understand. Also, there needs to be a line drawn as to if the user is being treated as a person with rights or if the user is being used as a source of data for the company's profit. When it comes to the companies and engineers, they have to consider if collecting so much of their user's personal data is a violation of their individual privacy rights. Additionally, they need to consider limiting their data collection even if it limits their profits.

Deontology has many strengths. It is a line of thinking that protects an individual's general rights and their right to privacy. Also, it is a framework that lets the individual establish ethical standards and responsibilities for themselves. Lastly, it prevents the individual from only making decisions that lead to a positive outcome. There are also a few weaknesses that come with deontology. This framework can become very complex in situations that involve tradeoffs that infringe on one person's rights but not another person's. Additionally, this framework does not always have clear rules or guidelines when duties of an individual conflict with each other.

Consequentialism

The ethical framework of consequentialism is a line of thinking that evaluates a decision based on the outcome of that decision. This means that a decision is considered ethical if it has an

outcome that has the greatest benefit possible while mitigating harm. This framework is especially relevant to this case because the potential consequences of smart home devices are very alarming. However, currently, the overall benefits of smart home devices have outweighed the risk of smart home devices. These devices give people benefits like convenience, automation, and accessibility. Furthermore, the data collected by the companies allows them to personalize the user's experience with the product, use the data to develop more advanced artificial intelligence systems, and improve other systems like speech recognition and control systems. Even though these benefits have been outweighing the consequences, the potential consequences are still there and need to be taken into account.

The privacy risks of these devices are still very prevalent. Even though the outcomes of large-scale data collection seem to have been very beneficial, the data can still be used against the users in a bad way. Data leaks and cyberattacks happen all of the time, when will it happen for these smart home devices? Also, there could be implications for how normalizing constant data collection inside of homes can harm society in the future.

Consequentialism has a few different strengths. First of all, it is a line of thinking that can help evaluate how large-scale events could affect society before they happen. Also, as I said before, it works well when analyzing problems that involve technology. Finally, it is a framework that analyzes both the positive and negative outcomes of a situation. There are also a few weaknesses within this framework. Even if there are good and bad outcomes to a situation, the decision would be made based on how beneficial the good decision is. It may not consider the bad outcomes enough. Also, this line of thinking may be hard to use when trying to predict long-term outcomes of a situation. Lastly, the framework will put individual rights to the side if society benefits as a whole from a decision.

Justification

These ethical frameworks are justified in how they will help analyze this case study. Consequentialism will help identify the pros and cons of smart devices within homes and their potential implications. Deontology provides a point of view from each stakeholder that shows how each one thinks and why they need these devices and their data. Using both of these ethical frameworks will provide an analysis that asks what is more important, the benefits that these devices bring to society, or the potential risk of an individual's privacy?

Analysis

This section will be used to identify the main issues that this case presents, analyze them using ethical frameworks, and provide solutions or options to those issues.

Ethical Issues

Informed Consent

The issue of informed consent has a few fundamental aspects to it. The biggest issue is that the companies that create these smart home devices use terms and conditions agreements that are very long and often hard to understand. Since most people do not want to read these long agreements, they just skip to the bottom of the page and agree. However, within these agreements, the user technically agrees with the company's data collection activities. Many of these customers are unaware of how long the data is stored, how much of the data is actually collected, and how much of the data is shared with other companies.

From a deontological point of view, the users have a right to make decision about their personal data that they are informed of. The topic of consent is brought into question due to users not being able to reasonably understand these agreements. On the other hand, these companies have a duty to communicate their terms honestly. Treating their users and customers as sources of data while knowing they don't understand the terms and conditions raises some serious concerns like violating the user's autonomy.

From a consequentialist point of view, both sides of the issue have different points. The companies will argue that collecting so much data improves the device's functionality and performance, the development of artificial intelligence integration, and the user's experience as a whole. These improvements would benefit millions of users that interact with the devices every day. On the contrary, a lack of transparency from the companies dissolves their trust with the public. Misusing this data can be harmful to society, which is a foundational piece of what consequentialism strives to avoid.

Surveillance and Privacy

Since these smart home devices are designed to constantly monitor the environment that they are in, questions about surveillance and privacy are raised. These devices are meant to be personalized to the user, so they record and store the user's voice for later use. This data has the potential to be misused due to sensitive data being recorded and stored.

From a deontological perspective, privacy is a fundamental right for each person. With this in mind, a person's home is generally considered their private space or area. Since the device is always monitoring the user, it brings up a potential invasion of privacy for the user. Additionally, since the device is designed by the engineers and their company, they have a duty to respect an individual's right to their privacy and minimize any unnecessary intrusion on the user.

From a consequentialist perspective, collecting data improves the device's functionality and performance. This makes it so society benefits from the convenience of the device and the technological advancement implications that it provides. On the other hand, so much surveillance

in a place of privacy being normalized could change a person's expectations of privacy. This large-scale monitoring could harm society as a whole with this normalization.

Data Collection and Misuse

This ethical issue involves data collection used to harm the public. Since so much data is being collected, it is vulnerable to cyberattacks and leaks that can expose sensitive user information. This exposure can lead to criminals stealing information to commit crimes like identity theft, financial fraud, and extortion.

From a deontological perspective, these companies have an obligation to protect their user's information and not let it get stolen. Allowing this data to be stolen violates this obligation and violates the ethical framework of deontology. The user should not be the one in harm's way due to the company's negligence.

On the consequentialist side, collecting so much data allows the improvement of the device's functionality and efficiency. However, since this data can be stolen it causes potential harm to the user. This is a fundamental flaw from a consequentialist perspective. Even further, the scale of this potential harm increases with an increase in data collection.

Solutions

Use Current Industry Practices

The first proposed solution is to maintain the industry's current practices and standards. This may not be the fanciest solution, but it is still one at the end of the day. This solution would entail continuing the current amount of data collection within smart home devices along with the same terms of service given to each user.

From a consequentialist point of view, this solution has both benefits and harms. The benefits are that the innovation continues. This means that the artificial intelligence systems will continue to improve along with the convenience and functionality. As a result, this will make the cost lower for consumers. On the other hand, the previously mentioned privacy concerns will still persist. Users will still be vulnerable to cyberattacks and hacks that can steal their data without them knowing. This can lead to the public being harmed and public distrust in the companies.

On the deontological side, the obligations remain the same from the company and its engineers. The informed consent will still remain a problem where the user autonomy is an issue. Companies should have an obligation to their users to not let their data be stolen, but they may still prioritize profits above them.

Transparency Increase and Explicit Consent

Another solution that should be considered is an increase in transparency and a mandate for opt-in consent for data collection. This would simplify the privacy agreements and require the

device's users to opt-in to data collection continuously. The privacy agreements would clearly state what data is being collected and why. This would include who has access to the data, what they are using it for, and how long it is being stored.

On the consequentialist side, the benefits of this solution are that public trust would increase and the ethical issues would be eliminated. This would lead to no public harm being done, a key point in consequentialism. However, this new change could lead to fewer devices being sold because of the transparency. Consumers may not want so much of their data being shared with random people or companies. This would lead to less data being collected, leading to innovation slowing down. Obviously, the fewer devices being sold would lead to the companies making less money as a result.

From a deontological point of view, the newly formed informed consent is a huge win. This ethical framework focuses on a person's obligations, and this solution fulfills the company's obligations to their consumers. Moreover, this solution makes it so the ethical obligations of honesty and transparency are met.

Data Minimization

This solution entails the devices only collecting essential data that allows them to operate. This would reduce the amount of data that is being stored and transferred to other, third-party companies that use the data for their own use. This essential data would still include voice commands and audio recordings; however, the device has to be turned ON for it to start "listening" for the user's command. The essential data would also include common types of data like video from security cameras, usage patterns and activity logs, and location information from mobile apps (MidSouth Fiber).

From a consequentialist perspective, the benefits would be attractive. The new data minimization policies would lower the risk of security breaches due to the lower data traffic being collected. This would lead to improved privacy protection and improve the public trust with the companies. However, the minimization would lead to the device having weaker features like personalization and functionality. This could lead to the technological advancement taking much longer than it currently is.

On the deontological side, the newly added data policies would increase the respect for the user's privacy rights. This is attractive to this ethical framework because it is a key point within it. Additionally, it would reduce the amount of potential intrusion on the user's life and makes it so potential harm to the public is diminished.

Recommendation

My recommendation consists of a combination of including mandatory, opt-in consent aspects for data collection and companies only collecting essential data from their users. The companies creating these products need to be more transparent and tell their consumers exactly what data is being collected, who all has access to it, and how long it is being stored for. Also, these companies need an opt-in requirement for their data collection so that the users are actively knowing that their data is being collected. This solution would limit the amount of data being collected along with third-party companies being able to use the data as often.

This solution is very appealing to the consequentialist ethical framework. The solution preserves the benefits of the smart devices like personalization and convenience, and the users can still choose to share more data if they want the more advanced features applied to their device. The greater transparency with the public would increase public relations and most likely continue the adoption of smart home devices in homes across the country. Furthermore, the data minimization would resolve conflicts like the security breaches and the concerns about surveillance.

On the deontological side, this solution makes it so the user's rights are better respected. This includes rights like informed decision-making, their individual privacy, and their autonomy. The data minimization policy would make it so there is a reduction in intrusion into the user's privacy. Additionally, the engineer and their companies would fulfill their ethical obligations to protect their user's rights by protecting their information and being transparent. Contrary to what was being done, the users would now be treated as individuals instead of sources of data.

This solution comes with many strengths. First of all, it creates a balance of ethical responsibility and innovation for the product. The smart home devices still have the ability to improve their functionality and the companies are innovating through ethical practices. Also, the solution would improve public trust and the public's perception of the companies that design these products. Furthermore, privacy and security risks would be diminished. Less data collection and informed consent mean less of a chance for cyberattacks to occur. Lastly, this solution still allows the companies to innovate upon their products. They will still gather data that will allow them to make improvements and changes to their product.

With a lot of strengths comes a few weaknesses. One main weakness is that, due to users opting out of the advanced data collection, companies could collect less data, slowing the device's development. This could make the device less personable to the user, making it less convenient. Furthermore, even if the terms of service is simplified, users may still ignore the updated privacy information, causing them to still be giving more data than they think. Lastly, a more broad weakness of this solution is that it would be difficult to enforce these standards consistently across all smart home device companies.

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